

Notes and DDS Frequency Words Reference

This table provides the mapping between musical notes (C0 to C8), their fundamental frequencies, and the corresponding 24-bit **freqWord** values required for the Oscillator DDS engine.

System Parameters:

- Phase Update Rate: 480 kHz
- Phase Width: 24 bits
- Tuning: A4 = 440 Hz

Note	Freq (Hz)	Hex	Dec	Note	Freq (Hz)	Hex	Dec
C0	16.35	0x00023B	571	C4	261.63	0x0023B2	9138
C#0	17.32	0x00025D	605	C#4	277.18	0x0025CD	9677
D0	18.35	0x000281	641	D4	293.66	0x00280D	10253
D#0	19.45	0x0002A7	679	D#4	311.13	0x002A76	10870
E0	20.60	0x0002D0	720	E4	329.63	0x002D05	11525
F0	21.83	0x0002FB	763	F4	349.23	0x002FBA	12218
F#0	23.12	0x000328	808	F#4	369.99	0x00328E	12942
G0	24.50	0x000358	856	G4	392.00	0x00358A	13706
G#0	25.96	0x00038B	907	G#4	415.30	0x0038B4	14516
A0	27.50	0x0003C1	961	A4	440.00	0x003C13	15379
A#0	29.14	0x0003FA	1018	A#4	466.16	0x003FA7	16295
B0	30.87	0x000436	1078	B4	493.88	0x004368	17256
C1	32.70	0x000476	1142	C5	523.25	0x004764	18276
C#1	34.65	0x0004B9	1209	C#5	554.37	0x004B99	19353
D1	36.71	0x000502	1282	D5	587.33	0x00501B	20507
D#1	38.89	0x00054F	1359	D#5	622.25	0x0054EC	21740
E1	41.20	0x0005A1	1441	E5	659.26	0x005A0B	23051
F1	43.65	0x0005F7	1527	F5	698.46	0x005F73	24435
F#1	46.25	0x000652	1618	F#5	739.99	0x00651D	25885
G1	49.00	0x0006B1	1713	G5	783.99	0x006B14	27412
G#1	51.91	0x000717	1815	G#5	830.61	0x007168	29032
A1	55.00	0x000783	1923	A5	880.00	0x007827	30759
A#1	58.27	0x0007F5	2037	A#5	932.33	0x007F4E	32590
B1	61.74	0x00086D	2157	B5	987.77	0x0086CF	34511
C2	65.41	0x0008ED	2285	C6	1046.50	0x008EC9	36553
C#2	69.30	0x000973	2419	C#6	1108.73	0x009733	38707
D2	73.42	0x000A03	2563	D6	1174.66	0x00A035	41013
D#2	77.78	0x000A9D	2717	D#6	1244.51	0x00A9D8	43480
E2	82.41	0x000B41	2881	E6	1318.51	0x00B416	46102
F2	87.31	0x000BEE	3054	F6	1396.91	0x00BEE7	48871
F#2	92.50	0x000CA4	3236	F#6	1479.98	0x00CA39	51769
G2	98.00	0x000D63	3427	G6	1567.98	0x00D629	54825
G#2	103.83	0x000E2D	3629	G#6	1661.22	0x00E2D1	58065
A2	110.00	0x000F05	3845	A6	1760.00	0x00F04D	61517
A#2	116.54	0x000FEA	4074	A#6	1864.66	0x00FE9D	65181

Note	Freq (Hz)	Hex	Dec	Note	Freq (Hz)	Hex	Dec
B2	123.47	0x0010DA	4314	B6	1975.53	0x010D9F	69023
C3	130.81	0x0011D9	4569	C7	2093.00	0x011D91	73105
C#3	138.59	0x0012E6	4838	C#7	2217.46	0x012E66	77414
D3	146.83	0x001407	5127	D7	2349.32	0x01406B	82027
D#3	155.56	0x00153B	5435	D#7	2489.02	0x0153B1	86961
E3	164.81	0x001683	5763	E7	2637.02	0x01682B	92203
F3	174.61	0x0017DD	6109	F7	2793.83	0x017DCE	97742
F#3	185.00	0x001947	6471	F#7	2959.96	0x019472	103538
G3	196.00	0x001AC5	6853	G7	3135.96	0x01AC51	109649
G#3	207.65	0x001C5A	7258	G#7	3322.44	0x01C5A1	116129
A3	220.00	0x001E0A	7690	A7	3520.00	0x01E09A	123034
A#3	233.08	0x001FD3	8147	A#7	3729.31	0x01FD3A	130362
B3	246.94	0x0021B4	8628	B7	3951.07	0x021B3F	138047
				C8	4186.01	0x023B23	146211